Bully and Ring Election Algorithms Performance Analysis Considering Real Case Aspects

Denise Comincioli, Alessandra Dalla Verde

Abstract—In the distributed systems field, various papers compared the performance of modified classic election algorithms with their original counterpart, by using their computational complexity to represent the number of messages exchanged. Currently, there is no analysis of classic algorithms' simulations, conducted under their theoretical assumptions or in real case scenarios. Hence, we decided to simulate the Ring and Bully election algorithms in the presence of communication delays and unreliable links. We compare their turnaround time and explore how its value changes in relation to a series of different factors (e.g. number of nodes). Finally, we analyze advantages and disadvantages of deploying the Bully and the Ring algorithm, before concluding that application requirements and network capabilities determine which one is the best for the distributed system at hand.

Index Terms—Election, simulation, performance evaluation, distributed algorithms, link failures

1 Introduction

In a distributed system having a coordinator node to manage operations is common practice. When the coordinator becomes unavailable or crashes, a new one must be selected to keep the system operating. This is achieved by using election algorithms, which guarantee consensus of all active processes on the new coordinator.
The Bully [1] and the Ring algorithm [2] are two classic election procedures. They have strong assumptions difficult to satisfy in a real system, where failures may occur and communication delays are a certainty. This paper aims to compare their performance based on turnaround times [1], exposing critical issues that may arise when breaking one or more assumptions.
In the following sections, we discuss: constraints and implementation choices, simulation core concepts, the performance analysis process and the most interesting simulations' final results.

2 Assumptions

The constraints we set for the Bully and Ring algorithms are extracted from academic literature and combined with some additional conditions. This integration was done to simplify implementation and allow for a better, fairer comparison of the two algorithms.
To build our set of assumptions, we referred to: the *Elections in a Distributed Computing System* paper by Garcia-Molina [1] for the Bully Algorithm, and the *Distributed Systems* book by van Steen and Tanenbaum [3], containing a simplification of Chang and Roberts' original work [2], for the Ring Algorithm. To summarize, we list our main assumptions for the final system:

- Distinguishable nodes: each node has a unique ID; IDs follow a total order.
- Closed system: nodes know all their peers; no processes can leave or enter the network.
- Asynchronous system.
- Reliable nodes: aside from the initial coordinator crash, no other node can fail during the election.
- Fully connected topology: every node is directly connected to all others.

According to the literature, the Ring algorithm requires nodes to be arranged in a logical ring, i.e., an overlay network. Nevertheless, we included the last point to allow more flexibility and achieve a common network architecture for both algorithms. This topology change does not impact the Ring algorithm's performance, as the nodes are still organized in a virtual ring based on their IDs.
Before proceeding with the implementation choices, we delve into the nature of the networks links we considered and the asynchronicity assumption.

2.1 Reliable and Unreliable Links

In the literature, the original formulation of the algorithms required the links to be reliable, meaning all messages were assumed to always arrive to the intended destination correct. In a real system, reliable links are expensive and difficult to achieve: different types of failures have to be managed, ranging from transmission errors to packet losses. Hence, we decided to analyze how the Ring and Bully algorithms performed when accounting for packet loss, in addition to measuring how they fared in the reliable case. When testing with unreliable links, both algorithms require the system to be synchronous.

2.2 Asynchronicity

An asynchronous distributed system allows no assumptions about message propagation delays, clock drift rates and process execution speeds [4]. Following this definition, the Bully algorithm breaks in an asynchronous system. A node becomes the new coordinator only when peers with a higher ID do not respond to the election messages it sent. This means that a process cannot simply wait for a response for

1. The terms turnaround time and runtime are used interchangeably throughout this paper.

an unbounded amount of time, as it might never receive one. Every node would need to know exactly the maximum round-trip-time to ensure the algorithm eventually terminates and is correct.
We minimized the issue by setting a maximum wait time that could cover the majority of the network delays generated. As a consequence, the performance of the Bully algorithm worsens, i.e., the turnaround time increases. Nevertheless, we can still collect valuable data from these simulations. The Ring algorithm under the reliable links assumption, does not present the same problem.
On the other hand, as discussed in Section 2.1, when considering unreliable links both procedures require a synchronous system. This stems from the fact that nodes need to distinguish lost from slow messages. Setting maximum timeouts is the easiest solution, as mistakenly retransmitting a slow message does not effect the outcome of the election process. We further cover this topic in Section 3.4.

3 Implementation Choices

This section explains how we implemented the two algorithms under the assumptions of reliable and unreliable links. In addition, it lists the factors that may influence the simulations' results and explains how we set them.

3.1 Algorithms Implementation

The Bully and the Ring algorithm rely on SimPy, "a process-based discrete-event simulation framework based on standard Python" [5], which allows us to manage process concurrency without threads, locks, or other low-level constructs. We chose to use this library to focus more on the implementation of the two algorithms.
In SimPy time is unitless, but developers can decide to give it the unit of measurement they prefer, as described in Section 3.2.2 of the SimPy documentation [6]. For this project we chose to use milliseconds, therefore, the performance results are expressed in this unit.

3.2 Unreliable Links Implementation

Aside from the alterations already discussed in Section 2.2, we had to implement more robust versions of the original algorithms to handle unreliable links. We integrated a fault tolerance technique to guarantee termination and, for the Bully algorithm only, correctness. In particular, we chose to adopt the retransmission strategy. It works as follows: a node sends a message, sets a timeout and waits for the acknowledgment from the receiver. If the ACK does not arrive before the timeout, the original sender retransmits the packet. ACKs can be lost, but the election final result remains correct even if a node receives a duplicated message. Performance-wise, this strategy causes the number of messages sent to increase for both the Bully and Ring algorithm.
In our simulation, packets are considered as lost if a randomly generated number ranging from 0 to 1 is lesser or equal than a loss rate parameter.

3.3 Initiators

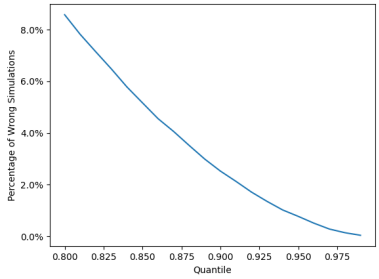
The original Ring and Bully algorithms accept a variable number of initiators, i.e., nodes which detect the coordinator's crash and start the election. We decided to keep this feature for all the algorithms' versions.

3.4 Delays and Timeouts

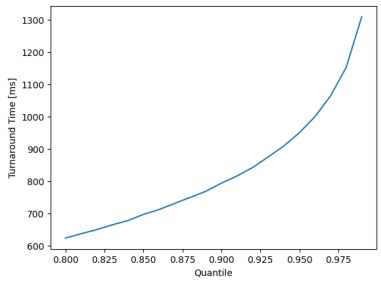
To simulate more realistic scenarios, we decided to generate random message delays. The paper *A Probabilistic Approach to Distributed Clock Synchronization* by Flaviu [7] addresses the issue of unpredictable communication delays in distributed systems and models them following a log-normal distribution. Therefore, in our system, we adopted a distribution with a similar trend for delay generation: the exponential. We chose it as it allowed for easier data extrapolation given its dependency on only one parameter, the rate, instead of two, like in the log-normal case.
In our algorithms, timeouts are set based on the maximum message delay possible to avoid breaking the election procedure. However, given the exponential distribution of the delays, arbitrarily large values can be generated, even if with low probability, meaning there is not a real maximum wait time. Hence, we computed timeouts based on a quantile parameter.
As discussed in Section 2.2, the Bully algorithm under the reliable links assumption requires timeouts to guarantee that eventually the active node with highest ID will be elected as the new coordinator. Regardless of the timeout value we set, the probability of a node missing a slow response and causing the wrong process to be elected, cannot be reduced to zero. Therefore, when testing under these assumptions, we set the quantile parameter to 0.99, which guarantees that more than 99% of the simulations ran are correct, as seen in *figure 1a*. The consequence of choosing higher quantile values is highlighted by *figure 1b*, where we can observe an increase of the election turnaround time.
To collect the data shown in *figure 1*, we ran the Bully algorithm 1000 times, counted the number of invalid simulations and stored the mean turnaround time. We then repeated this process 500 times for each quantile. In this case, we used a simulation environment with the following parameters: network size of 5 nodes, 1 election initiator and a delay mean of 110 ms.
When considering unreliable links, we set for both algorithms a quantile parameter of 0.8, as the retransmission of messages prevents these type of issues.

3.5 Number of Nodes

"Many leader election algorithms apply to only relatively small distributed systems" [8]. To verify that this statement holds also for the Bully and Ring algorithms, we performed simulations with the same parameters on networks of different size, where the number of nodes ranged from 3 to 25. We set the other environment parameters, i.e., number of initiators and delay mean, equal to the ones used to collect data for the quantile graphs of Section 3.4. For each type of simulation, we run the two algorithms 10000 times, storing the mean of the turnaround time and the mean of the number of messages. Based on these metrics, we can observe

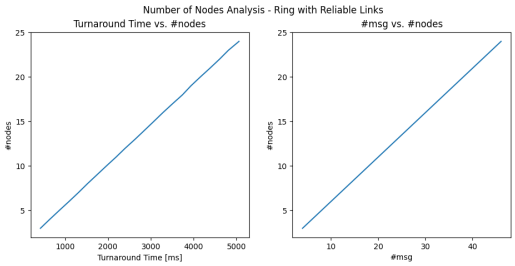


(a) Percentage of Wrong Simulations

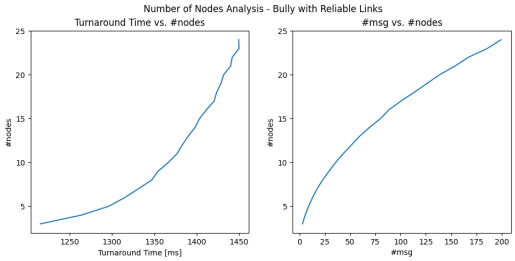


(b) Turnaround Time

Fig. 1: Exponential Quantile Analysis of Reliable Bully Algorithm



(a)



(b)

Fig. 2: Number of Nodes vs. Number of Messages and Turnaround Time

that the performance of the two algorithms decreases as the number of nodes grows, which is highlighted by *figure 2*. In particular, *figure 2a* shows that the number of messages and the turnaround time of the Ring algorithm increase linearly with the number of nodes. The number of messages may be affordable, but not the turnaround time. Meanwhile, we can see from *figure 2b* that the Bully algorithm runtime grows slower, but the number of messages increases rapidly. If a distributed system needs to avoid traffic caused by only control messages, this last algorithm is not recommended. A similar argument can be made with the versions of both procedures for the unreliable links assumption and is provided in the Appendix (see *figure 8*).
Based on the results presented, we decided to simulate the two algorithms with small networks.

4 Performance Analysis

We evaluated the performance of the two election algorithms based on two metrics: the number of messages sent and the turnaround time, i.e., the time the network takes to reach consensus on the new coordinator.
We created different testing scenarios by changing the following parameters: number of nodes, loss rate and number of initiators. We kept the mean of the exponential delay distribution fixed at 110 ms.
In total we tested networks with 4, 5, 6 and 8 nodes in scenarios with: 20%, 50% or 75% loss rate, and 1, 3 or 4 election initiators. For convenience, in later sections we use the words "*default scenario*" to refer to cases in which: the network has 5 nodes, there is only one initiator and, if under the unreliable link assumption, the loss rate is set to 20%.
We computed the two performance metrics by applying the same procedure used in Sections 3.4 and 3.5: to achieve reliable data, we repeated the execution of each algorithm 10000 times. We decided to exclude the outliers by following the IQR method, before computing the sample variance, the sample mean, and the mean 95%-confidence interval.

5 Results

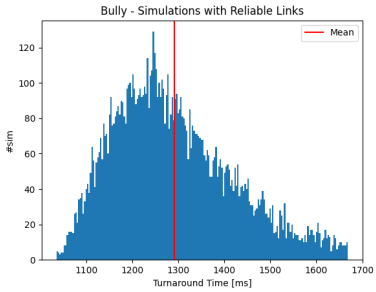
This section shows the results we achieved from the simulations. First, we analyze the performance of the two algorithms separately, then we compare them.

5.1 Bully Algorithm Analysis

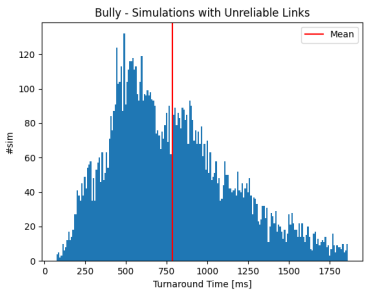
Before proceeding with version-specific observations, we offer some insights on behaviors shared by both the reliable and unreliable links versions of the Bully algorithm. As expected, a higher number of nodes causes an increase of both the turnaround time and the number of messages. Additionally, in similar simulations, a higher initiator count corresponds to a runtime decrease and a message increase. The latter phenomenon is explained by the fact that, even if later interrupted, at least in the beginning each initiator separately starts their own election. On the other hand, the algorithm turnaround time improves because the more initiators are involved, the higher are the chances to produce an election where the number of nodes to contact and the generated delays are lower. Essentially, the fastest conducted election eventually suppresses all others.

5.1.1 Reliable Links

When considering the *default scenario*, under the reliable links assumption, the Bully algorithm has a mean turnaround time of 1290.83 ± 2.61 ms and a 17234.51 ms^2 variance. As previously discussed, the quantile parameter was set to 0.99. The presence of high variance in the data is explained by the random choice of the initiator: if the starting node is already the one with highest ID, the election



(a)



(b)

Fig. 3: Bully Algorithm Turnaround Time Analysis

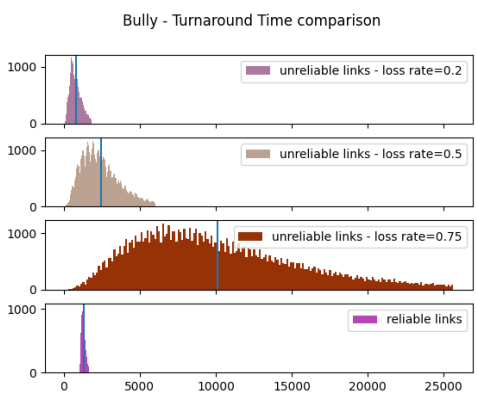
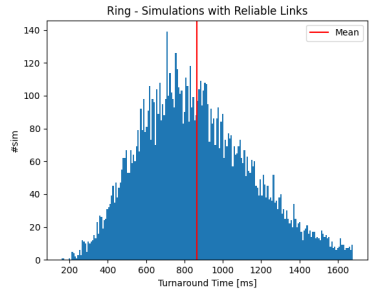


Fig. 4: Bully Algorithm Turnaround Time Comparison with Reliable and Unreliable links

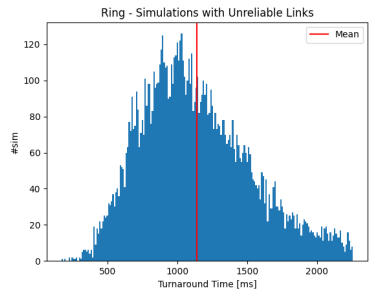
terminates almost immediately, while if it has the lowest ID and the generated message delays are particularly high, the procedure will take longer.
In *figure 3a*, we can see that the turnaround time distribution skews to the left, indicating that it is more common for the algorithm to have lower runtimes.

5.1.2 Unreliable Links

In the *default scenario* under the unreliable links assumption, the Bully algorithm has a mean turnaround time of 786.02 ± 7.38 ms and a 137489.76 ms^2 variance.
Against expectations, it might appear that the algorithm performs better with unreliable links, but it is not as it seems. We already highlighted in Section 3.4 that the quantile parameter used for this case is 0.8, therefore all maximum wait times are reduced compared to the reliable version of the algorithm. Only when the number of messages lost is



(a)



(b)

Fig. 5: Ring Algorithm Turnaround Time Analysis

relatively low, i.e., when the loss rate is less than approximately 45%, the reduced wait times can offset the cost of retransmitting packets. When the loss rate exceeds 45%, the Bully algorithm performs worse when under the unreliable links assumption. We can see this phenomenon in *figure 4*.

5.2 Ring Algorithm Analysis

The general considerations made in Section 5.1 are also valid for the Ring algorithm: the turnaround time decreases as the number of initiators increases. The Ring algorithm stops when an initiator receives back their message after it completed two round trips of the ring. Hence, smaller communication delays cause the election started by one initiator to win over the others, affecting the final turnaround time and increasing the number of messages sent.

5.2.1 Reliable Links

The reliable Ring algorithm, in the *default scenario*, has a turnaround time mean of 862.88 ± 5.68 ms and a turnaround time variance of 82618.75 ms^2.
As in the case of the reliable Bully algorithm, the variance is very high, but it is only attributed to the randomization of the exponential delays. Due to the fact that messages are exchanged only between two neighbors, and the election does not proceed before the receiver has processed the message, the performance of the algorithm is strictly influenced by how much time a message takes to be transmitted.
In *figure 5a* we present the histogram of the reliable Ring algorithm turnaround time in the *default scenario*. It behaves like a Gamma distribution, as we are summing 10000 Gamma-distributed samples, each obtained as the sum of random exponential sample values.

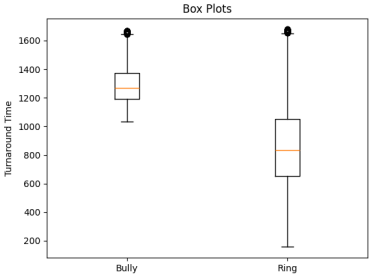


Fig. 6: Turnaround Time Box Plots of Ring and Bully Algorithms

5.2.2 Unreliable Links

The Ring algorithm with unreliable links, in the *default scenario*, has a turnaround time mean of 1137.66 ± 7.84 ms and a turnaround time variance of 156166.45 ms^2. Compared to the reliable links version, the performance worsens due to packets loss. The considerations we made about the distribution in the previous section also hold in this case, but we do not have an exact Gamma distribution, as we see in *figure 5b*. This is caused by the variability of the number of messages.

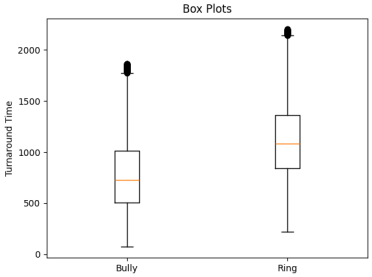
5.3 Bully and Ring Algorithms Comparison

5.3.1 Reliable Links

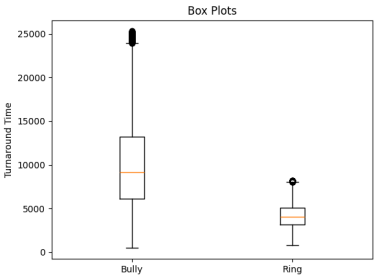
In *figure 6* we show the turnaround time box plots for the Bully and Ring algorithms with reliable links. The simulations we ran used the *default scenario* parameters. Due to the quantile set for timeouts, the Ring algorithm performs better than the Bully. Specifically, the former has lower turnaround times because it does not rely on timeouts, and therefore is not penalized by them, as already mentioned in section 5.1.1. We do not show other additional plots, because, even when considering other testing scenarios, the relation between the Ring and Bully algorithms remains the same. This changes only when analyzing networks with more than 8 nodes, but as discussed in Section 3.5, we are not interested in doing so.

5.3.2 Unreliable Links

With unreliable links, based on the chosen loss rate, the relation between the Ring and Bully algorithms changes significantly. In the *default scenario* (*figure 7a*), the Bully algorithm performs better than the Ring. Unlike the reliable case, here the Ring algorithm is also penalized by timeouts. Furthermore, when a packet is lost during the Ring election, the messages exchange is stuck between two neighbors. Until the receiver processes the packet, the transmitter must continue to send the lost message to the same node. Instead, in the Bully algorithm, multiple processes can simultaneously send messages to other nodes. Hence, even if some packets are lost, nodes can receive information from different transmitters. This ensures a faster election.
With loss rates greater than approximately 45%, the Ring algorithm wins over the Bully algorithm. *Figure 7b* shows this phenomenon with a loss rate of 75%. Packets losses tend to penalize more the Bully algorithm, as explained in Section 5.1.2.



(a) Loss Rate of 0.2



(b) Loss Rate of 0.75

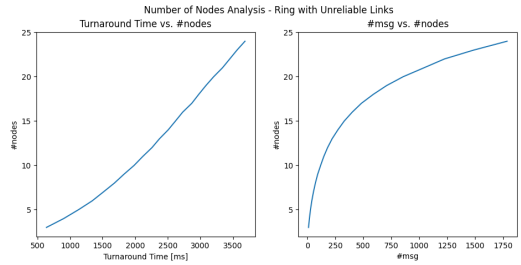
Fig. 7: Turnaround Time Box Plots of Ring and Bully Algorithms with Unreliable Links

6 Conclusion

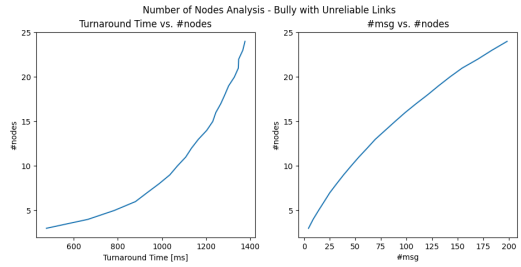
After conducting our analysis, we can determine that there is no absolute best performing election algorithm between the two. In general, having more initiators decreases turnaround time, but a threshold should be determined to avoid clogging the channels. When considering reliable links, it is clear that applying a Bully algorithm requires more adjustments due to the costly network assumptions needed to make it work as intended. Therefore in this case the simplicity of the Ring algorithm allows for better performance and more predictability. These aspects can be very valuable to reduce election related traffic in a small distributed system. Nevertheless, the Ring algorithm has a hidden operational cost related to maintaining the virtual ring structure when the topology changes. In practice, where networks do not normally reach a 45% loss rate, the situation is reversed and the unreliable links version of the Bully algorithm appears to be the best candidate. To conclude, our paper presented interesting and relevant aspects of applying two of the simplest election algorithms known. For a more complete evaluation, our work should be integrated with studies accounting for process failures.

Appendix
Number of Nodes Analysis under Unreliable Links

As the reliable links case, the performance of the Ring and Bully algorithms under unreliable links decreases as the number of nodes grows. *Figure 8* shows this result under the same conditions and with the same procedure described in Section 3.5. The most relevant aspect regards the Ring algorithm. Observing the number of messages (*figure 8a*),



(a)



(b)

Fig. 8: Number of Nodes vs. Number of Messages and Turnaround Time with Unreliable Links

the control traffic generated is difficult to manage.
Instead, the Bully algorithm with unreliable links (*figure 8b*) is pretty similar to the reliable links case.

References

[1] H. Garcia-Molina, *Elections in a Distributed Computing System*, IEEE Tr. on Comp., Vol. C-31, NO. 1, Jan. 1982.

[2] E. J. H. Chang and R. Roberts, *An improved algorithm for decentralized extremal finding in circular configurations of processes*, Communications of the ACM, vol. 22, no. 5, pp. 281–283, 1979.

[3] M. van Steen and A. S. Tanenbaum, *Distributed Systems*, 4th ed., Version 4.03, Jan. 2025, ch. 5, sec. 5.4.2.

[4] G. Coulouris, J. Dollimore, T. Kindberg, and G. Blair, *Distributed Systems: Concepts and Design*, 5th ed. Boston, MA: Addison-Wesley, 2012, ch. 2, sec. 2.4.1.

[5] Team SimPy, *SimPy Documentation*, Release 4.0.0, Apr. 8, 2020.

[6] Team SimPy, *SimPy Documentation*, Release 4.0.0, Apr. 8, 2020, ch.3, sec. 3.2.2.

[7] F. Cristian, *A probabilistic approach to distributed clock synchronization*, in Proc. 9th Int. Conf. Distrib. Comput. Syst., 1989, pp. 288–296.

[8] M. van Steen and A. S. Tanenbaum, *Distributed Systems*, 4th ed., Version 4.03, Jan. 2025, ch. 5, sec. 5.4.5.